

Occupancy Grid & EKF SLAM

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Occupancy Grid

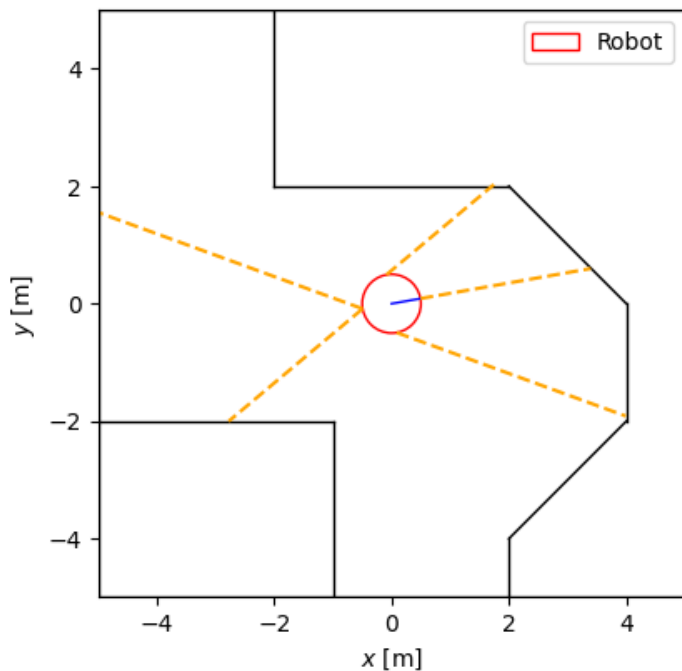
The concept behind the occupancy grid is fairly easy and straight forward. The likelihood of a cell is updated additively based on the inverse sensor model.

```
1: Algorithm occupancy_grid_mapping ( $\{\ell_{t-1,i}\}, x_t, z_t$ ):  
2: for all cells  $m_i$  do  
3:   if  $m_i$  in perceptual field of  $z_t$   
4:      $\ell_{t,i} = \ell_{t-1,i} + \text{inv\_sensor\_model}(m_i, x_t, z_t) - \ell_0$   
5:   else  
6:      $\ell_{t,i} = \ell_{t-1,i}$   
7:   endif  
8: endfor  
9: return  $\ell_{t,i}$ 
```

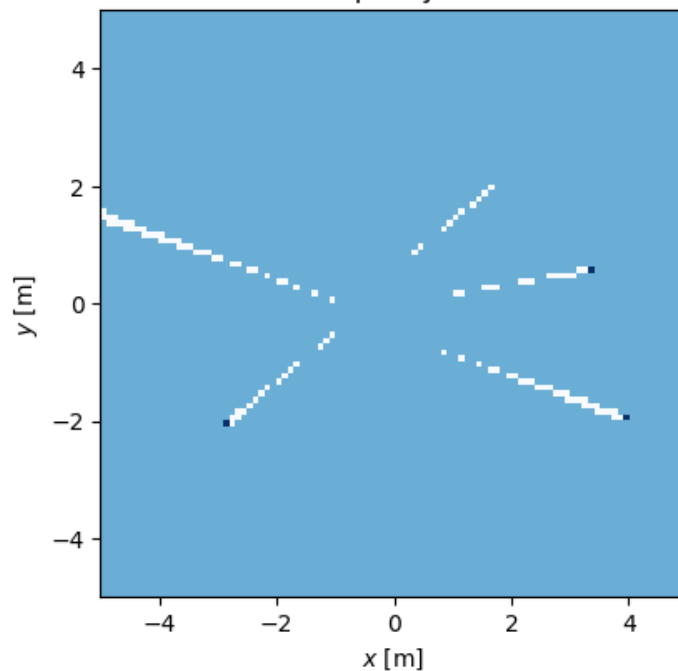
Occupancy Grid – Initial Cond.

Initial State

Robot with Measurements w.r.t. the Groundtruth



Occupancy Grid

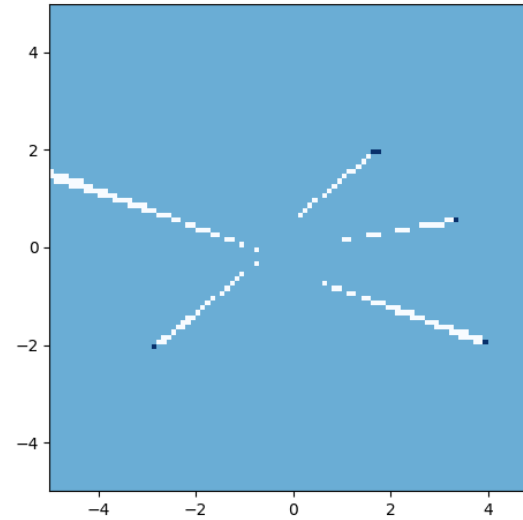
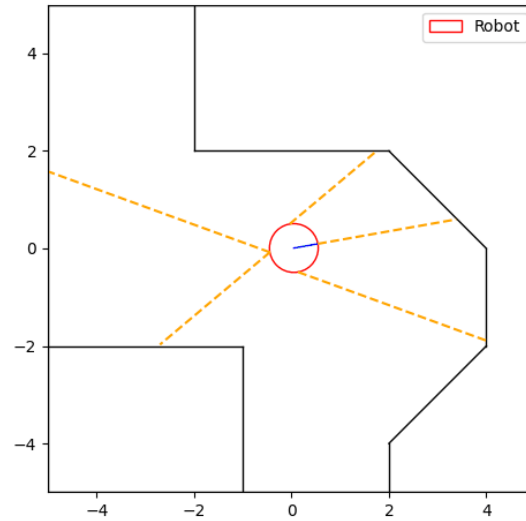


Occupancy Grid – The Simulation

The simulation runs with a sensor that implements collision avoidance, and the mapping occurs at every measurement.

The represented Occupancy Grid is thresholded to the closest of the discrete values: {Free, Occupied, Unexplored}

Occupancy Grid - Simulation



EKF

Much like the Kalman Filter, the EKF SLAM algorithm uses the same theoretical fundamentals and can be divided into:

- **Prediction Step:** $\bar{\mu}_t = g(u_t, \mu_{t-1})$
 $Q_t \bar{\Sigma}_t = G_t \Sigma_t G_t^T + F_x^T R F_x$
- **Correction Step:** $K_i^t = \bar{\Sigma}_t H_i^T (H_i \bar{\Sigma}_t H_i^T + Q_t)^{-1}$
 $\mu_t = \bar{\mu}_t + K_i^t (z_i^t - \hat{z}_i^t)$
 $\Sigma_t = (I - K_i^t H_i) \bar{\Sigma}_t$

EKF SLAM's Extended State

The EKF SLAM works with an extended state μ and Σ , which opposed to the KF for pose estimation, also includes estimates with their respective uncertainties of the landmarks. This yields the following kalman variables to be updated:

$$\mu = [\mu_x \quad \mu_m]^T = [\mu_x \quad \mu_m^{(0)} \quad \dots \quad \mu_m^{(N-1)}]^T$$
$$\Sigma = \begin{bmatrix} \Sigma_{xx} & \Sigma_{xm} \\ \Sigma_{mx} & \Sigma_{mm} \end{bmatrix} = \begin{bmatrix} \Sigma_{xx} & \Sigma_{x,m_0} & \dots & \Sigma_{x,m_{N-1}} \\ \Sigma_{m_0,x} & \Sigma_{m_0,m_0} & \dots & \Sigma_{m_0,m_{N-1}} \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma_{m_{N-1},x} & \Sigma_{m_{N-1},m_0} & \dots & \Sigma_{m_{N-1},m_{N-1}} \end{bmatrix}$$

EKF SLAM Prediction Step

The Prediction Step updates the position estimate of the robot, this step takes into account:

- The update of the robot's position estimate (μ_x, Σ_{xx})
- The update of the cross-correlations for the uncertainties of the landmark position w.r.t. the robots position Σ_{xm}

Note that all other elements of the extended state μ and Σ will not be influenced by this first step

EKF Prediction Step

Indeed, if expanding the expressions of the prediction step, yields the following equations for the prediction step:

$$\bar{\mu}_x^t = g(u_t, \mu_{t-1})$$

$$\bar{\Sigma}_{xx}^t = J_t(\mu_x) \Sigma_{xx}^t J_t(\mu_x)^T + R$$

$$\bar{\Sigma}_{mx}^t = \Sigma_{mx}^t J_t(\mu_x)^T$$

EKF SLAM Correction Step

The Correction Step updates the estimate with its uncertainties by means of the measurement. Expanding the formulas of the EKF we get:

- The innovation Ψ
- The Kalman Gain K

The Innovation

The innovation Ψ can be computed with a limited set of computations, and does not require the brute force multiplication. The selector matrices F influence only specific regions of the concerned states and uncertainties:

$$\Psi = H_i \bar{\Sigma}_t H_i^T + Q_t = H_x \Sigma_{xx} H_x^T + H_m \Sigma_{m_i, x} H_x^T + H_x \Sigma_{x, m_i} H_m^T + H_m \Sigma_{m_i, m_i} H_m^T + Q_t$$

The Kalman Gain

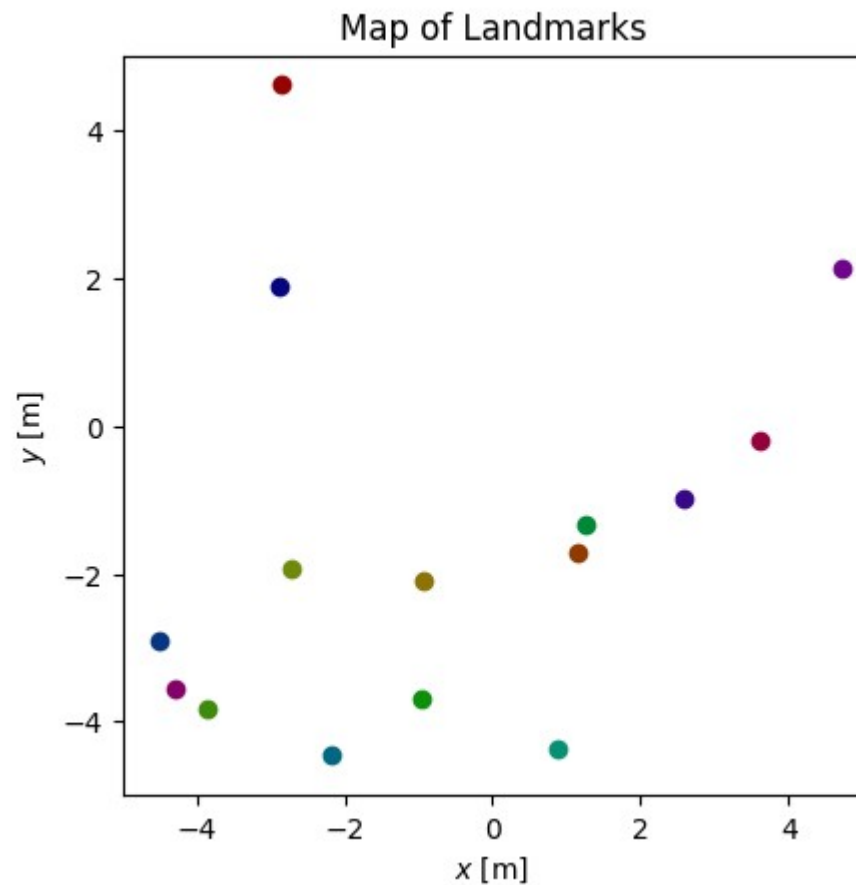
The Kalman Gain, similarly can be computed with a limited set of operations, instead of implementing the multiplication of the full dimensions elements:

$$K_i = \begin{bmatrix} K_x^{(i)} & K_0^{(i)} & \dots & K_i^{(i)} & \dots & K_{N-1}^{(i)} \end{bmatrix}^T$$

$$K_x^{(i)} = (\Sigma_{xx} H_x^T + \Sigma_{xm_i} H_m^T) \Psi^{-1}$$

$$K_k^{(i)} = (\Sigma_{m_k x} H_x^T + \Sigma_{m_k m_i} H_m^T) \Psi^{-1}$$

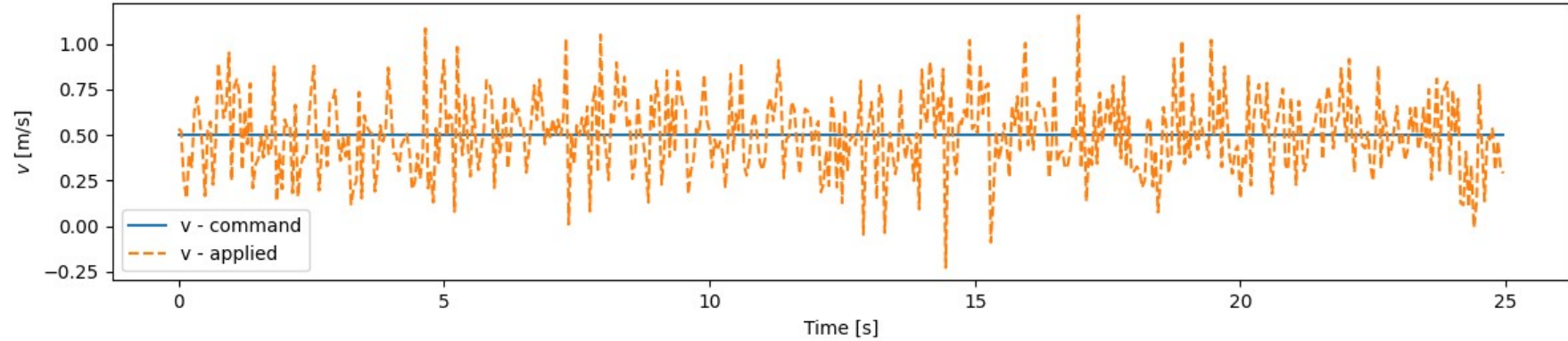
The Simulation - Map



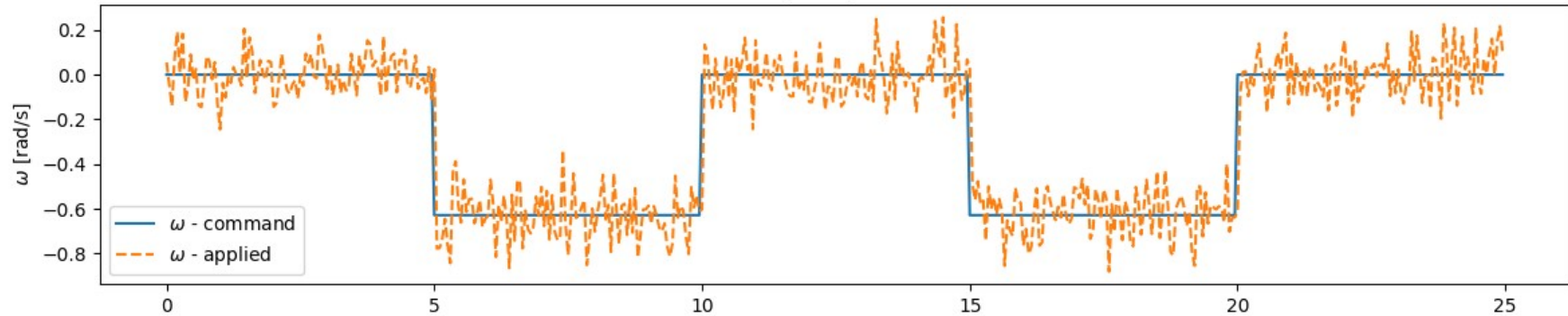
The Simulation - Trajectory

Applied VS Command Trajectories

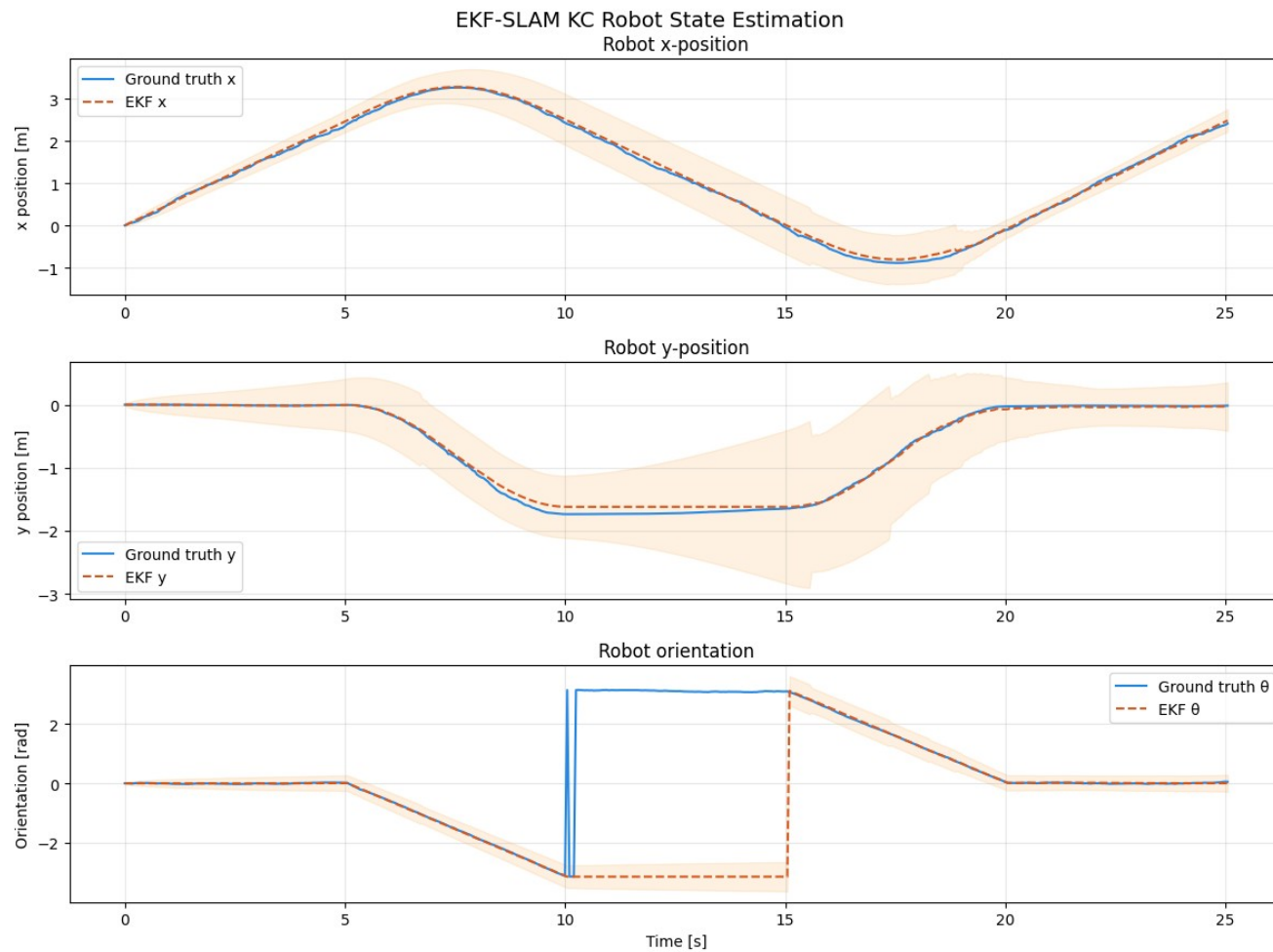
Trajectory for v



Trajectory for ω



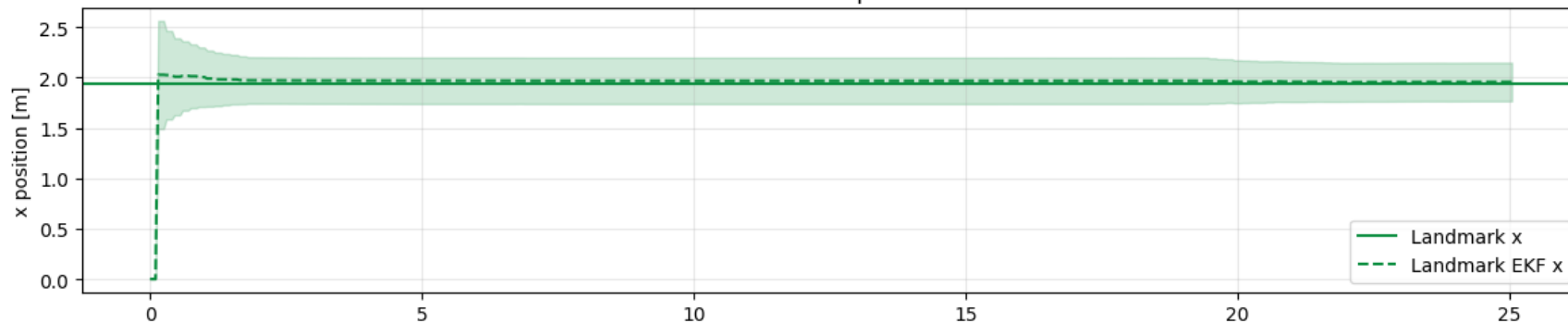
The Simulation – Pose Est.



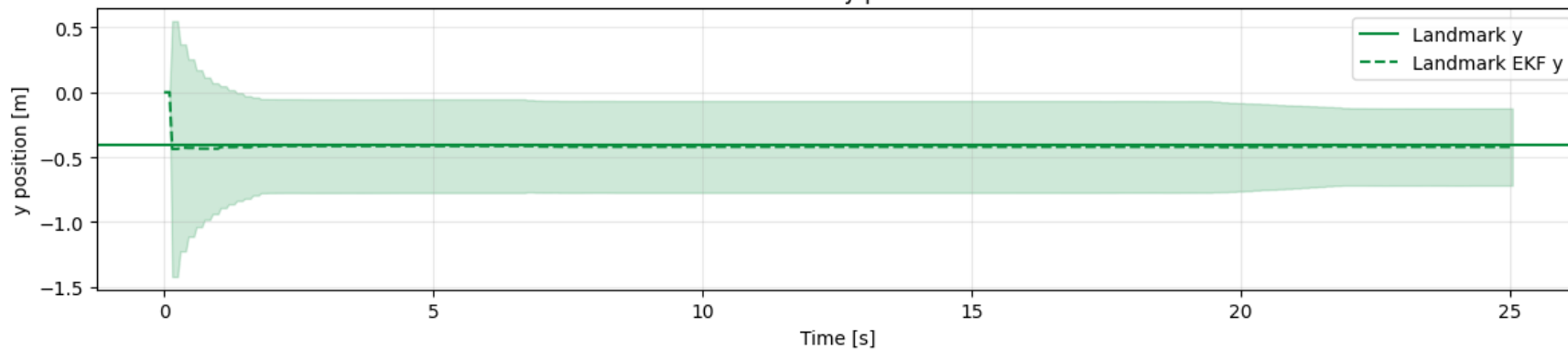
The Simulation – LM Pose Est

EKF-SLAM KC Landmark 0 Estimation

Landmark x-position



Landmark y-position



About α

Alpha is the threshold that influences the acceptance of new landmarks.

- High values of alpha tend to reduce the numbers of newly added landmarks, while
- lower numbers tend to increase the number of new landmarks.

Indeed, if too low, several landmark estimates are created for the same landmark in the GT, conversly, too high alphas map all measurements to a reduced and incomplete set of landmark estimates.